

Legal name (U.S.): Daniel Han Kim

CONTACT INFORMATION

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RESEARCH STATEMENT

EDUCATION

March 2013–February 2019

PUBLICATIONS

- C1. Aiyappa, R., DeVerna, M. R., Pote, M., Truong, B. T., Zhao, W., Axelrod, D., Pessianzadeh, A., Kachwala, Z., Kim, M., Seckin, O. C., **Kim, Minsuk**, Gandhi, S., Manikonda, A., Pierri, F., Menczer, F. & Yang, K.-C. *A multi-platform collection of social media posts about the 2022 us midterm elections in Proceedings of the international AAAI conference on web and social media* **17** (2023), 981–989.

Under Review

- R1. Weng, G., **Kim, Minsuk**, Ahn, Y.-Y. & Moro, E. Beyond Distance: Mobility Neural Embeddings Reveal Visible and Invisible Barriers in Urban Space. *arXiv:2506.24061* (2025).

HONORS AND AWARDS

Fellowship

- U.S. Department of the Air Force Summer Faculty Fellowship Program, 2024
- Luddy Doctoral Summer Fellowship, Indiana University, 2023, 2025

Scholarship

- Scholarship of Dean of Graduate School, Kyung Hee University, 2019
- Superiority Scholarship, Kyung Hee University, 2017
- Exemplary Scholarship, Kyung Hee University, 2013, 2014

Conference

- Best Oral Presentation Award, The 21st Workshop for Statistical Physics (Online), 2021
- Best Poster Presentation Award, The Korean Physical Society Spring Meeting, (Online), 2020
- Best Oral Presentation Award, The 20th Workshop for Statistical Physics (Byeonsan, Korea), 2019

Miscellaneous

- Best Project Award, The 19th KIAS-APCTP Winter School on Statistical Physics, (Pohang, Korea), 2022
- Best Project Award, The 17th KIAS-APCTP Winter School on Statistical Physics, (Pohang, Korea), 2020
- Excellence Project Award (3rd place), The 10th KIAS CAC Summer School on Scientific Computing & Artificial Intelligence, (Seoul, Korea), 2019

TEACHING

Indiana University

- Associate Instructor
 - Performance Analytics (INFO-I369) Spring 2025, Fall 2025

Kyung Hee University

- Guest Lecturer
 - Advanced Statistical Physics (PHYS7037) November 15th, 2021
title: “Finite-size scaling analysis in Statistical Physics”
 - Statistical Mechanics 2 (PHYS7072) December 14th, 2020
title: “Statistical Physics of Opinion Dynamics”
- Teaching Assistant
 - Modern Physics (PHYS2309) Spring 2019, Spring 2020
 - Computational Physics (PHYS3308) Spring 2019, Spring 2020
 - Informations for Physics (PHYS3310) Fall 2019, Fall 2020

MENTORING

M.S. and post-undergraduate students

- Bilal Sultan (Data Science, Indiana University) 2025-present

Undergraduate students

- Bilal Sultan (Data Science, Indiana University)

2024-2025

PRESENTATIONS

Conference talks

- NetSci 2025, "Minimum-cost percolation on US air transportation network" (Maastricht, Netherlands, June, 2025)
- NetSci 2025 Satellite Symposium: Critical Phenomena In Networks, "Shortest-path percolation on complex networks" (Maastricht, Netherlands, June, 2025)
- IC2S2 2024, "Unsupervised embedding of mobility reveals invisible barriers in US cities" (Philadelphia, PA, USA, July, 2024)
- NetSci 2024, "Shortest-path percolation on complex networks" (Québec City, Canada, June, 2024)
- APS March Meeting, "Shortest-path percolation on complex networks" (Minneapolis, MN, USA, March, 2024)
- The 21st Workshop for Statistical Physics, "Control of epidemic spreading on layered networks" (Online, August, 2021)
- KPS Spring Meeting, "Effective epidemic control model on multilayer networks" (Online, April, 2021)
- The 20th Workshop for Statistical Physics, "Degree-weighted Majority-vote model on Complex Networks" (Byeonsan, Korea, August, 2019)

Conference posters

- NetMob 2024 data challenge, "Investigating Human Mobility Patterns with Field Theory and Open Data Sources" (Washington, D.C., USA, October, 2024)
- KPS Spring Meeting, "Impact of degree-weighted influence in the Majority-vote dynamics on Complex Networks" (Online, July, 2020)
- KPS Fall Meeting, "Majority-vote model with degree-weighted influence on Complex Networks" (Gwangju, Korea, October, 2019)

Colloquia and Seminars

- Department of Physics, Kyung Hee University, "Shortest-path percolation on complex networks", (Seoul, Korea, December 20th, 2024)
- **Complex Systems Research Exchange (CREx)**, "Shortest-path percolation on complex networks", (Online, January 30th, 2024)
- Kyung Hee University Academic Society Seminar: Scaling and Theoretical Physics, "Finite-size scaling analysis in Statistical Physics" (Seoul, Korea, November 26th, 2021)
- Kyung Hee University Theoretical physics research group seminar, "Majority-vote model with degree influence on complex networks" (Seoul, Korea, May 22nd, 2020)

SERVICE

Reviewer

- Journal: Communications Physics, IEEE Transaction on Network Science and Engineering, Nature Communications, Physical Review E, Scientific Reports
- Conference: NetSci (2026, 2025), IC2S2 (2025, 2024)

SKILLS

Programming Language: C, Python, R

Relevant tools: NetworkX, Gephi, OSMnx, Geopandas, statsmodels

Specialization: Monte Carlo simulation, Finite-size scaling analysis

Language: Korean (native), English (fluent)

REFERENCES

Dr. Filippo Radicchi

Professor

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Dr. Yong-Yeol Ahn

Quantitative Foundation Distinguished Professor

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Dr. Esteban Moro

Professor

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Dr. Santo Fortunato

James H. Rudy Professor of Informatics and Computing

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Dr. Soon-Hyung Yook

Professor

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